

Model Card: ViT-DeCV

1. Model Details

- **Model Name:** ViT-DeCV (Vision Transformer with Decorrelated Attention Head Output Vector)
- **Model Type:** Image Classification Model (Fine-tuned Vision Transformer)
- **Version:** 1.0
- **Base Architectures:**
 - ViT-L-14-laion2B-s32B-b82K (Large)
 - google/vit-base-patch16-224-in21k (Base)
- **Architecture & Design:**
 - **Core Mechanism:** Incorporates a **Diversity Loss (L_{div})** term, which penalizes the cosine similarity between attention head output vectors in the final layer.
 - **Loss Function:** $L_{total} = L_{classification} + \lambda \cdot L_{div}$
 - **Calculation:** L_{div} is computed as the Frobenius norm of the difference between the cosine similarity matrix (S) and the identity matrix (I), using $ReLU$ to focus only on positive correlations: $L_{div} = ||ReLU(S - I)||_F^2$
 - **Training Strategy:** Utilizes a partial fine-tuning strategy (Defreeze), freezing early layers and training only the last layer to conserve computational resources. For ViT-B-32, this results in approximately 7.5M trainable parameters.
- **Training Framework & Hardware:** Trained using PyTorch; fine-tuning ViT-L-14 takes approximately 2 hours for 10 epochs on an A100 GPU (80GB memory).

2. Training Data

The model was evaluated on three datasets to test generalization, stability, and robustness:

- **CIFAR-10:** A 10-class dataset used for initial validation and generalization testing.
- **CIFAR-100:** A 100-class dataset used to evaluate training stability and performance on more complex tasks.
- **Waterbirds (Sagawa et al., 2019):**
 - **Purpose:** Specifically designed to test robustness against **spurious correlations**.
 - **Composition:** Combines bird types (Waterbirds/Landbirds) with conflicting backgrounds (Water/Land). It includes "worst-group" scenarios where the background does not match the bird (e.g., landbird on water).

3. Performance Metrics

Performance is evaluated using **Top-1 Accuracy** and **Standard Deviation (SD)** to measure stability across multiple training runs.

- **Training Stability (CIFAR-100):**

- **Baseline:** Accuracy 83.75%, SD 0.06.
 - **ViT-DeCV (MyModel):** Accuracy 86.83%, **SD 0.001**.
 - **Analysis:** The model demonstrated a 3.08% accuracy gain and a drastic reduction in standard deviation (from 0.06 to 0.001), proving functions effectively as a stabilizer.
- **Generalization (CIFAR-10):**
 - **Baseline:** Accuracy 95.95%.
 - **ViT-DeCV:** Accuracy 96.85% (+0.9%).
 - **Analysis:** Improvement was marginal because the pre-trained baseline already achieves high accuracy and inherent diversity.
- **Robustness & Worst-Group Performance (Waterbirds - ViT-B-32):**
 - **Baseline:** Full Accuracy 82.08%.
 - **ViT-DeCV:** Full Accuracy 83.62% (+1.54%).
 - **Worst-group (0,1) Accuracy:** Improved from 70.17% to 73.61%.

4. Intended Use

- **Primary Use Cases:**
 - Tasks requiring **high training stability** and reproducibility across random seeds.
 - **Resource-constrained fine-tuning** where full-model training is not feasible, and only the final layers can be updated.
- **Out of Scope:**
 - Tasks where the baseline model is already highly optimized and diverse (high-baseline), as L_{div} yields limited gains.
 - Tasks requiring deep semantic disentanglement where statistical orthogonality is insufficient compared to semantic interventions.

5. Limitations and Failure Modes

- **Statistical Independence $\neq$ Semantic Diversity:**
 - **Limitation:** Enforcing statistical orthogonality via L_{div} does not guarantee that the model learns semantically "meaningful" or distinct features.
 - **Evidence:** In ViT-L-14 Waterbirds experiments, L_{div} regularization actually resulted in slightly lower accuracy (91.75%) compared to the fine-tuned baseline (92.27%).
- **Natural Diversity:** Pre-trained ViT models inherently develop diverse attention heads during standard fine-tuning (even with $\lambda=0$), which limits the effectiveness of explicitly adding L_{div} .
- **Computational Cost:** Fine-tuning large models (e.g., ViT-L-14) remains resource-intensive despite freezing early layers.

6. Fairness and Bias

- **Bias Mitigation:** The model was evaluated on its ability to handle **spurious correlations** (e.g., relying on background instead of the object). On ViT-B-32, ViT-

DeCV improved performance on the "worst-group" (mismatched background), suggesting reduced reliance on background cues.

- **Comparison with Ablation:**
 - While L_{div} helps, **User-Guided Ablation** (manually removing heads correlated with background features via CLIP analysis) proved significantly more effective.
 - **Data:** On ViT-L-14, User-Guided Ablation achieved 93.73% accuracy, outperforming L_{div} 's 91.75%.
 - **Conclusion:** For addressing specific biases, understanding and controlling the **semantic roles** of attention heads is more effective than blind statistical decorrelation.

7. Ethical Considerations

- **Research Prototype:** This model is a research prototype designed to explore regularization techniques for Attention mechanisms.
- **False Sense of Fairness:** Users should not rely solely on statistical regularization (L_{div}) to ensure model fairness. While it improves stability, experiments show it is less effective than targeted semantic interventions for eliminating deep-seated biases (like background dependency). Deployment in sensitive contexts requires rigorous semantic verification.